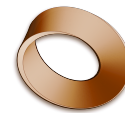


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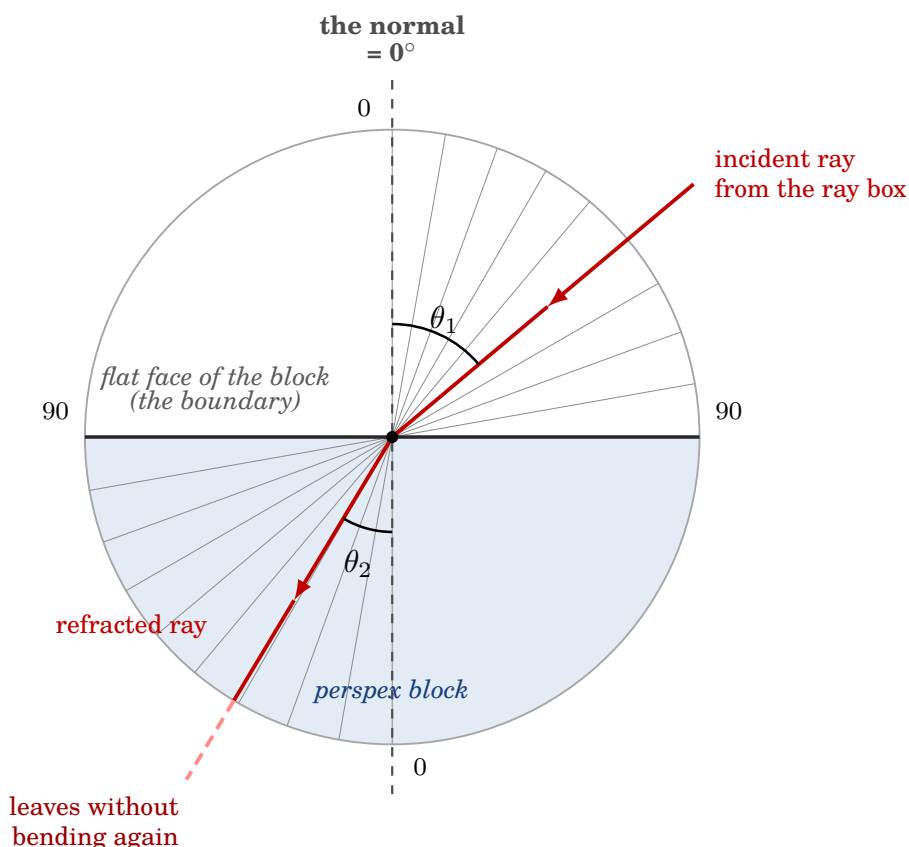
1.5 – Measuring Refraction

Unit 1: Light & EM Spectrum

Learning intentions

Last lesson we marched the boundary ourselves and found out **why** light bends. Today we find out **how much** it bends — and we learn the way of measuring it that you will need for your investigation.

Your block is a **semicircular** piece of perspex, and it sits on a printed card with **two protractors** on it. The block goes in the blue half, with its **flat face** along the line across the middle. You aim the ray at the **centre** of the card — and then you read both angles straight off the card. No tracing. No lifting.



Three things to notice on that diagram before you touch any apparatus.

- The ray bends when it crosses the **flat face** — that is the boundary between air and perspex. It bends **towards the normal**, because perspex slows light down.

- **The normal is the dashed vertical line, and on this card it is 0°.** Both angles are measured from it.

- On the way out, the ray leaves through the **curved** face — and it does **not** bend again. Question 3(e) asks you why.

1. The method

You have a **ray box**, a **semicircular perspex block**, and the **printed protractor card**. You will not need paper, a pencil trace, or your own protractor — the card is the protractor.

1. Put the block on the card in the **blue half**, curved side down. Line its **flat face** up exactly along the line across the middle, with the middle of the flat face on the **centre** of the card.
2. Switch the ray box on and make a **single** narrow ray. Aim it so that it hits the **centre** of the card — the point where the two protractors meet. Check this **every time**, before every reading.
3. Swing the ray box round until the incident ray lies along the 0° line of the top protractor. That is your first reading: an angle of incidence of 0° .
4. Read where the refracted ray comes out on the **bottom** protractor. Write it in the table.
5. Swing the ray box round to 10° , then 20° , and so on up to 80° , reading the bottom protractor each time.

The two instructions printed on your card

“Set up your ray box so that the light ray hits the **centre** of the protractors every time you take a reading.”

“Remember to measure from the **normal** (90° from the surface it is hitting), and take **the normal as 0°** .”

Finding the right scale — do this first

Your card has **two** scales printed on it. The one you want is the one that reads **0° at the dashed normal**, counting up to 90° along the flat face.

Here is the check. Do step 3 — send the ray straight down the normal. **The scale you want reads 0°** . Find that scale once, at the start, and every reading you take afterwards is measured from the right place.

Safety. The ray box gets **hot**. Do not touch the bulb, and keep paper, sleeves and pencil cases out of it. If you smell burning, switch it off at the power supply and tell Mr. Rawson.

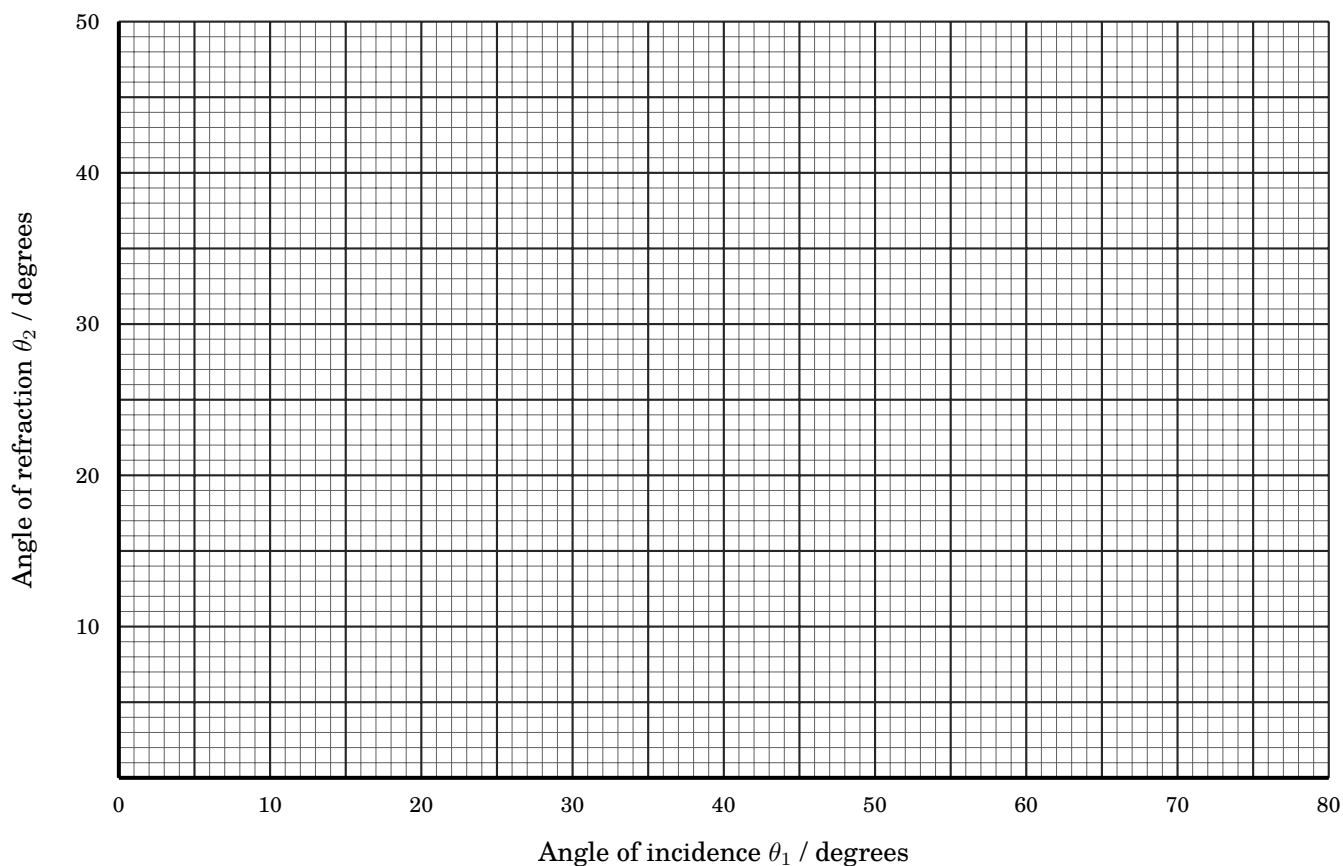
2. Your results

Read the angle of refraction to the **nearest degree**. The third column is just the first two subtracted — it tells you *how much* the ray bent.

Angle of incidence θ_1 , in air	Angle of refraction θ_2 , in perspex	How much it bent $\theta_1 - \theta_2$
0°		
10°		
20°		
30°		
40°		
50°		
60°		
70°		
80°		

3. Draw the graph

Plot **angle of refraction** (up) against **angle of incidence** (across). Plot your nine points with a sharp pencil, then join them with one smooth curve.



4. What your results show

(a) Compare each pair of angles. Is θ_2 always bigger than θ_1 , always smaller, or does it vary?

(b) Use the word **speed** to explain why.

- (c) What would happen to θ_2 if your block were made of **diamond** instead of perspex?
($n_{\text{perspex}} = 1.49$, $n_{\text{diamond}} = 2.42$)

- (d) Look at your 0° row. The ray came in **along the normal**. What did it do, and why?

- (e) The ray leaves the block through the **curved** face, and it does not bend there. Use the word **normal** to explain why not.

*Hint: the centre of the card is the centre of the circle, so the ray inside the block is travelling straight out along a **radius**.*

Extension

For each row of your table, work out $\theta_1 \div \theta_2$.

Then work out $\sin \theta_1 \div \sin \theta_2$.

One of those two calculations gives roughly the **same answer every time**. Which one — and what is that number close to?
